Technology Stack

Date	03 July 2026
Team ID	xxxxxx
Project Name	**OptiCrop: Smart Agricultural Production Optimization Engine**
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	**Architecture Component / Layer**	**Technology Chosen**	**Justification / Purpose**
1	**Frontend / Client-Side**	HTML5, CSS3, JavaScript	Developed a responsive and user-friendly interface for farmers to enter soil parameters and view crop recommendations. HTML structures the webpage, CSS provides an attractive design, and JavaScript enables interactive validation and smooth user experience.

2	**Backend / Server-Side**	Python, Flask	Python is used for Machine Learning integration and data processing. Flask provides a lightweight web framework that connects the frontend with the trained Random Forest model and processes prediction requests efficiently.
3	**Database / Data Storage**	CSV Dataset (Current Version) Future: MongoDB	The trained model was built using an agricultural crop dataset stored in CSV format. MongoDB is planned for future versions to store prediction history, user information, and recommendation logs.
4	**Cloud / Hosting / Deployment**	Local Flask Server Future: Render / Railway / AWS	The current project runs locally for development and testing. Future deployment on cloud platforms will provide online accessibility, scalability, and remote usage by farmers.
5	**Version Control & CI/CD**	Git, GitHub	Git tracks project changes, while GitHub stores the repository online, enables collaboration, version management, documentation, and project submission.

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools**	Scikit-learn NumPy Pandas Matplotlib Pickle VS Code Google Colab Future: OpenWeather API, SHAP Explainability	Justification / Purpose Scikit-learn – Train and evaluate the Random Forest model. NumPy – Numerical computations and array operations. Pandas – Data preprocessing and analysis. Matplotlib – Data visualization during model development. Pickle – Save and load the trained Machine Learning model. VS Code – Project development and debugging. Google Colab – Model training and experimentation. OpenWeather API (Future) – Real-time weather integration. SHAP (Future) – Explain why a crop was recommended.